

WAINWRIGHT CFB, AB, Canada

WMO: 711180

Lat: 52.8305N

Lon: 111.0957W

Elev: 686

StdP: 93.35

Time Zone: -7.00 (NAM)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -32.3	(c) -29.3	(d) -36.3	(e) 0.1	(f) -32.2	(g) -33.1	(h) 0.2	(i) -29.2	(j) 11.3	(k) -2.4	(l) 10.0	(m) -1.1	(n) 2.8	(o) 230	(p) 0.666

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 12.1	(c) 29.7	(d) 17.2	(e) 27.6	(f) 16.6	(g) 25.8	(h) 15.8	(i) 19.0	(j) 26.2	(k) 18.0	(l) 25.1	(m) 17.0	(n) 23.7	(o) 4.5	(p) 160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.7	12.9	21.4	15.5	12.0	20.4	14.4	11.1	19.6	56.9	25.8	53.4	25.1	50.3	23.7	23.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 9.9	(b) 8.5	(c) 7.5	(e) -36.4	(f) 33.8	(g) 3.3	(h) 2.3	(i) -38.8	(j) 35.5	(k) -40.7	(l) 36.9	(m) -42.6	(n) 38.2	(o) -45.0	(p) 39.9		
(a) 9.9	(b) 8.5	(c) 7.5	(e) -36.5	(f) 20.7	(g) 3.3	(h) 1.3	(i) -38.9	(j) 21.6	(k) -40.8	(l) 22.4	(m) -42.6	(n) 23.1	(o) -45.0	(p) 24.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.1	-12.7	-11.0	-4.9	4.2	10.8	15.0	18.0	16.8	11.9	4.4	-4.3	-11.5		
	DBStd	12.64	9.15	7.75	8.00	5.58	4.58	3.33	3.10	3.43	4.48	5.20	6.75	8.19		
	HDD10.0	3297	703	587	463	183	46	3	0	1	31	183	429	667		
	HDD18.3	5621	962	821	722	423	237	109	44	70	198	432	679	925		
	CDD10.0	786	0	0	0	11	69	153	247	211	87	9	0	0		
	CDD18.3	68	0	0	0	0	2	8	33	21	3	0	0	0		
	CDH23.3	1061	0	0	0	3	66	132	411	337	109	2	0	0		
	CDH26.7	283	0	0	0	0	11	27	121	94	30	0	0	0		
(14) Wind	WSAvg	3.6	3.5	3.4	3.7	4.1	3.9	3.5	3.3	3.2	3.5	3.7	3.5	3.4		
(15) Precipitation	PrecAvg	408	21	13	18	30	42	73	71	55	33	18	18	17		
	PrecMax	547	61	24	43	71	104	110	151	125	83	48	61	30		
	PrecMin	261	4	2	1	4	10	17	18	6	1	4	3	2		
	PrecStd	85	13	7	10	16	24	26	30	24	22	10	13	7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.8	5.8	13.7	22.8	28.1	29.3	32.6	32.8	30.4	22.7	12.5	7.5		
		MCWB	2.2	2.3	7.2	11.3	14.1	16.6	18.7	18.0	15.8	12.5	6.0	3.1		
	2%	DB	3.9	3.6	9.6	19.8	25.1	26.7	29.8	29.1	26.3	18.0	8.9	4.4		
		MCWB	1.1	0.7	4.1	9.5	12.8	15.6	18.1	17.2	14.5	10.3	4.3	1.4		
	5%	DB	1.9	1.6	6.7	16.5	22.6	24.6	27.6	27.0	23.6	15.1	6.3	1.7		
		MCWB	-0.3	-0.5	2.7	7.7	11.8	14.7	17.6	16.8	13.7	8.4	2.7	-0.6		
	10%	DB	-0.5	-0.3	4.5	13.7	20.0	22.4	25.4	24.9	20.6	12.3	4.0	-0.6		
		MCWB	-2.2	-2.0	1.4	6.3	10.5	13.8	16.7	16.0	12.4	6.7	1.3	-2.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.5	2.6	7.2	11.7	15.4	18.5	20.9	20.0	16.7	12.8	6.6	3.5		
		MCDB	5.3	5.6	13.3	21.6	24.3	25.8	27.2	26.9	25.6	21.5	11.4	7.1		
	2%	WB	1.2	1.0	4.6	9.9	13.7	16.9	19.5	18.7	15.3	10.7	4.7	1.4		
		MCDB	3.7	3.2	8.6	18.8	22.9	23.6	26.7	25.8	24.7	16.6	8.6	4.2		
	5%	WB	-0.2	-0.4	2.9	8.2	12.5	15.8	18.5	17.6	14.1	9.0	3.0	-0.5		
		MCDB	2.0	1.5	6.4	15.6	20.8	22.1	25.5	24.8	22.3	14.6	5.8	1.6		
	10%	WB	-2.2	-2.1	1.6	6.7	11.3	14.7	17.5	16.6	12.8	7.3	1.5	-2.3		
		MCDB	-0.5	-0.4	4.3	12.7	18.8	20.8	24.0	23.5	19.6	11.7	3.8	-0.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.2	9.8	9.6	11.5	12.5	11.3	12.1	12.7	12.8	10.7	8.6	8.7		
		MCDBR	10.1	11.1	12.0	16.4	16.4	15.3	15.9	17.2	18.5	15.7	11.3	10.3		
		MCWBR	8.1	9.0	7.8	8.5	7.3	6.8	6.6	7.1	8.3	8.6	7.7	8.1		
		MCDBR	10.2	10.9	11.3	15.2	14.8	12.8	13.9	14.6	16.8	14.7	10.8	10.3		
	5% WB	MCWBR	8.4	8.9	7.6	8.3	7.0	6.3	6.5	6.7	7.9	8.3	7.6	8.3		
(40) Clear-Sky Solar Irradiance	taub		0.254	0.295	0.290	0.333	0.354	0.371	0.364	0.378	0.317	0.280	0.256	0.234		
	taud		2.319	2.227	2.404	2.353	2.346	2.329	2.383	2.321	2.461	2.540	2.450	2.357		
	Ebn at Noon		754	809	909	901	896	880	880	844	868	842	759	722		
	Edh at Noon		58	88	91	109	116	119	111	112	86	64	51	46		
(44) All-Sky Solar Radiation	RadAvg		1.06	2.12	3.59	4.73	5.80	5.96	6.26	5.09	3.59	2.12	1.06	0.79		
	RadStd		0.10	0.16	0.27	0.44	0.40	0.42	0.32	0.40	0.38	0.29	0.11	0.07		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (72 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page